

Summary of On Evaluating Performance of LLM Inference Systems

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Problem and Motivation

Due to the rapid evolution of Large Language Models (LLMs), there exists a divergence between the system technology at unprecedented scale and the methods for evaluating them, which have remained relatively static and sometimes obscure true performance metrics, which impede the scientific progress of analysis of these systems. Existing evaluation methods are fundamentally flawed since they give rise to recurring anti-patterns in three key dimensions: *Baseline Fairness, Evaluation Setup, and Metric Design*.

The aforementioned anti-patterns are uniquely inherent to LLMs due to their dual-phase inference, which couples a compute-intensive parallel prefill phase for the prompt with a memory-bandwidth-bound sequential decode phase for token generation, creating complex resource contention. Furthermore, workload heterogeneity arises due to diversity in prompt length, output length, and latency requirements. The rapid architectural evolution of models is continually shifting performance bottlenecks.

The authors attempt to address this problem by formulating a comprehensive checklist aimed at providing guidance to ensure that experimental evaluations assess system performance. They establish a framework for recognizing and avoiding anti-patterns in favor of robust LLM inference evaluation. They also present a case study using speculative decoding to produce bursty, non-uniform token generation where naive evaluation approaches usually fail.

Related Works

The formulated conventional performance evaluation metrics for LLM inference, which are currently in use:

- **Time to First Token (TTFT)** - Used in interactive applications which have low latency requirements.
- **Time to last Token (TTLT)** - Used in applications where partial outputs have limited utility.
- **Time Between Tokens (TBT)** - Reflective of perceived model speed.
- **Time Per Output Token (TPOT)** - Decoding time.
- **Normalized Latency** - Used for throughput comparison.
- **Capacity** - Reflective of sustainable request load-bearing ability.

Solution Overview

The primary contribution of the paper is the introduction of a rigorous evaluation framework designed to identify and mitigate common anti-patterns in the performance evaluation of LLM inference systems. This framework consists of a detailed checklist that addresses three key areas:

1. **Baseline Fairness:** The framework emphasizes the importance of ensuring fair comparisons between systems. This involves avoiding comparisons to weak or overly simplified baselines and ensuring that performance improvements are correctly attributed to algorithmic advances, rather than just engineering optimizations (e.g., tuning hardware or using more efficient batch processing).
2. **Realistic Evaluation Setup:** A major focus of the framework is to ensure that the evaluation setup reflects real-world production environments. This includes designing evaluation workloads that mirror the unpredictable nature of interactive traffic, accounting for factors such as burstiness (rapid bursts of token generation) and user variability (e.g., requests of varying lengths and complexity). This setup ensures that the evaluation process simulates the conditions of live, user-facing systems as closely as possible.
3. **Meaningful Metric Design:** Rather than relying solely on simplified metrics like average latency or throughput, the paper proposes more comprehensive metrics that consider user experience. This includes factors like generation smoothness, variability in response times, and tail latency, metrics that are crucial for interactive tasks where users expect consistent and coherent outputs.

To demonstrate the application of this framework, the paper includes a case study of speculative decoding, a technique designed to improve generation speed by predicting and precomputing the next tokens in parallel. The case study highlights how conventional evaluation practices can mislead researchers. In this case, traditional metrics may incorrectly judge speculative decoding as ineffective or suboptimal because they fail to account for generation bursts (temporary increases in token generation rate) and tail latency (delays towards the end of generation), both of which are crucial for understanding the real-time performance of such systems.

Limitations

The limitations of this paper stem from its lack of wide acceptability as a standardised metric of evaluation. Firstly, the authors use speculative decoding which is one among many optimization techniques used in LLM inference, without being able to sufficiently justify it being a representative workload. This disallows it to be generalized across workloads.

Secondly, the authors employ a level of subjectivity in deciding on the key factors based on which they formulate anti-patterns. For example, there is ambiguity about what is considered “fair” for parameter tuning and which workloads are “representative” of larger inputs. This exacerbates the difficulty in adopting the comprehensive checklist provided by the authors as an industry standard.

Future Research Directions

Future research in this field will be decided by industry trends. Based on the depth of research, standards, benchmarks and baselines collectively accepted and widely used by academia. The aim of future work will be to reconcile the cost function associated with evaluation metrics which currently take into consideration primarily the latency and throughput. The cost could also comprise the associated energy requirements.

Summary of Class Discussion

Q. The paper discusses several anti-patterns, but how subjective are these in practice? Do the authors acknowledge this subjectivity or outline directions for future research? Are the identified anti-patterns sufficient for evaluating current LLM system designs?

Settling on a particular implementation framework remains challenging, and evaluating each system independently from the overall technique is often impractical. The validation process is long, resource-intensive, and slows the research process. Many papers are never fully *cleaned up* or re-implemented to check for all possible anti-patterns, leading to inconsistent rigor across experiments.

Moreover, evaluation practices in LLM systems research are becoming less standardized due to the constantly changing field. There is no universally accepted benchmark or workload, unlike in traditional domains such as databases. As the field stabilizes, a standardized benchmark suite will likely emerge. Overall, the field remains highly dynamic. Academia tends to focus on narrow, problem-specific optimizations that are difficult to integrate into real-world systems. The larger open challenge lies in combining these fragmented solutions to achieve consistent, end-to-end speedups across the entire LLM inference pipeline.

Q. In speculative decoding, the trade-off is between power consumption and model accuracy versus throughput. Why did the authors focus only on throughput and latency?

Throughput and latency are traditional, system-centric performance metrics in systems research. Energy efficiency is a relatively new consideration and is often reframed as cost. Most LLM system papers, therefore, prioritize throughput and latency, while power and cost metrics are only beginning to gain attention.

Summary of Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services

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Problem and Motivation

Large Language Models (LLMs) are increasingly used in text-streaming applications such as chatbots and virtual assistants, where the model generates tokens one by one to simulate a real-time conversation. However, existing LLM systems focus mainly on maximizing throughput or minimizing latency on the server side rather than optimizing the perceived user experience.

The currently used first-come-first-served (FCFS) scheduling often causes long delays even before the first token appears, frustrating users, while later tokens that are produced come much faster than humans can read, leading to wasted GPU cycles without improving satisfaction. This misalignment between system-level metrics and human perceived responsiveness is the fundamental problem the paper is trying to resolve by *Andes*. The motivation behind *Andes* is to redefine efficiency for such LLM-based text streaming by introducing a Quality of Experience (QoE) metric that captures the entire user interaction timeline to produce how natural the interaction feels to the user.

Related Works

Recent work on LLM inference serving has focused on boosting throughput, latency, and memory efficiency through systems like Orca, vLLM, Splitwise, and DistServe, but most rely on FCFS scheduling and overlook user experience. Scheduling approaches such as VTC and Llmunix improve fairness and load balancing yet ignore Quality-of-Experience (QoE). Unlike video streaming, where QoE depends on bandwidth-related factors, LLM-based text streaming is constrained by GPU compute and responsiveness. Andes addresses this gap by defining QoE as a core objective and introducing a preemptive scheduling algorithm that prioritizes responsiveness and coherence, shifting LLM optimization from raw performance to user-centric quality.

Solution Overview

The architecture follows a co-design approach between the LLM inference server and the client application to produce an optimal end-to-end user experience. The server focuses on when and how to allocate GPU resources among concurrent requests, while the client ensures that the user receives the responses at a smooth pace, which is perceivable with the user's reading or listening speed.

On the server side, Andes employs a QoE-aware request scheduler that operates at token granularity. Instead of processing requests to completion in a first-come-first-served (FCFS) manner, the scheduler can pause and resume requests dynamically between token generations. This fine-grained preemption allows Andes to significantly reduce time-to-first-token (TTFT) for waiting users while ensuring that active users continue receiving tokens rapidly enough to maintain seamless text streaming. To address challenges such as variable resource demand (arising from differences in prompt and response lengths), shared GPU utilization, and preemption overhead from saving and restoring key-value (KV) caches, Andes continuously monitors the real-time QoE and resource usage of all active requests. It prioritizes those whose QoE is at risk of degradation and dynamically reallocates GPU compute and memory. Andes manages the effects of enlarged batch sizes by adjusting how many and which requests share GPU cycles to balance throughput and smoothness.

On the client side, Andes uses a token pacer to regulate the rate at which tokens are displayed or vocalized. The pacer temporarily buffers excess tokens and releases them steadily at the user's consumption speed. This mechanism smooths out fluctuations caused by GPU scheduling, network jitter, or temporary preemptions. The server would be aware of the pacer's buffer status, generating only as many tokens as needed to keep the client's queue full to avoid excess computation while also guaranteeing continuous output.

The overall workflow proceeds as follows: the client submits a request along with its QoE parameters (desired TTFT and consumption speed). The server's Request Tracker records this metadata and forwards it to the token-level scheduler, which dynamically decides which requests to execute or pause, optimizing the aggregate QoE under GPU memory and compute constraints. Generated tokens are then immediately pushed to the client, where the token pacer ensures consistent, human-aligned delivery.

Limitations

1. The QoE Parameters that are used may vary dynamically or even be difficult to collect in real-world scenarios, which could affect scheduling accuracy and fairness.
2. There is a concern of synchronization complexity that can incur because of preemption overhead (context switching on a fine-grained token level) for large models with massive KV caches.

3. It is currently focused on single-node setups and does not extend to multi-server or distributed cluster environments.
4. Different users exhibit diverse reading and listening speeds, leading to varying QoE expectations that are difficult to maintain consistently across sessions.

Future Research Directions

Future research on Andes' preemptive scheduling approach could explore several directions to further optimize Quality of Experience (QoE) in LLM-based text streaming. These include expanding QoE metrics to capture additional factors, such as content relevance, and developing adaptive scheduling that tailors performance to user needs and session context. Scalability remains a key challenge, so future work could focus on optimizing the scheduling algorithm for high-concurrency environments and distributed systems. Additionally, integrating machine learning techniques like reinforcement learning could dynamically improve QoE in real-time. Finally, combining QoE optimization with system performance metrics and creating more robust evaluation frameworks for real-world user studies would help refine and generalize the approach across diverse applications.

Summary of Class Discussion

Q) Are there any practical experiments validating that the proposed QoE metric truly represents or validates real user experience?

Some aspects of user experience are inherently subjective, and the paper does not clearly demonstrate how well QoE correlates with actual human perception or satisfaction.

Q) How does Andes' token-level preemption interact with batch size and memory usage?

Batching improves GPU memory efficiency but introduces potential context-switching overhead when requests are frequently preempted. Andes accounts for this by factoring preemption cost into its scheduling priority.

Q) How would Andes apply to *diffusion-based* or *reasoning* language models, where text generation is not strictly linear?

While Andes assumes token streaming, a similar QoE framework could be adapted by redefining the atomic unit from tokens to blocks. For diffusion models, one could measure per-block delivery aligned with user consumption time, creating an analogous QoE for block-based generation.

The discussion concluded that Andes is best suited for interactive workloads where QoE is more meaningful, such as chatbots, assistants, and speech interfaces.